Dear Editors,

We submit a Research article, transferred through the OUP Transfer Desk after an editorial decision at Briefings in Bioinformatics (manuscript BIB-26-2439, declined before peer review without reviewer reports), entitled "Auditing cross-cohort feature-selection benchmarks: cohort composition and tissue-context-dependent confounding across kidney, liver and colon transcriptomes." This is the full-length version of that submission: every method and control analysis is in the main text, and the complete data tables and audit records are supplied as Additional files 1-7.

Why GigaScience. The article is built around reproducibility as a measured property rather than a claim, which is the criterion your journal states. It asks whether a cross-cohort feature-selection benchmark measures the methods compared or the cohorts used, and answers with three results that are each regenerable from source data:

1. A benchmark verdict can be a property of the cohorts. Over all 26 subsets of five diabetic kidney disease discovery cohorts, the reproducibility difference between a stability selector and a co-expression hub criterion ranged from -0.106 to +0.170 across three-cohort subsets, changing sign, while the mean external AUROC advantage stayed positive. The sweep was repeated under seven inner selectors and with the reference WGCNA package as the comparator.

2. A shared pre-analytical confounder becomes the most reproducible signal. Every discovery contrast compares needle biopsies with surgical tissue, and an immediate-early gene module tracked that contrast in 8 of 9 kidney datasets. A biopsy control retained 14-56% of the naive signal across seven diagnoses; residualising on a handling score removed most of the procurement-driven genes while batch correction by cohort was inert.

3. Both findings were tested in liver and colon against predictions fixed in advance. Three predictions were written and SHA-256-stamped before any liver or colon value was read; two failed, and the failures are reported in full. In colon, where both arms are biopsies, the module rose with inflammation, so the confounding it indexes depends on tissue context.

Research objects. The harmonised per-cohort matrices, the frozen 9,900-gene space, the sample-level phenotype and procurement annotations, and the cohort audit as a relational database whose views enforce the exclusion and overlap rules are deposited at https://doi.org/10.5281/zenodo.23051697, and the code at https://github.com/kdh6126/dkd-cross-cohort-audit (MIT). A single entry point regenerates every figure, table and reported number through 133 declared stages with fixed seeds, and a SHA-256 manifest ties the submitted files to that run. The registered cross-tissue predictions and their digests are in Additional file 7. All primary data are public GEO, KPMP, PRIDE, Mendeley Data, GWAS Catalog and Metabolomics Workbench records, listed with accessions in the Data availability section. Should the article be accepted, we will also deposit the processed data in GigaDB under CC0.

Journal policies. The study analysed only previously published, de-identified public data and generated no new human or animal data, so no ethics approval was required. The custom co-expression comparator is named as such throughout and is not presented as the reference WGCNA implementation, which was run separately. The manuscript is original, is not under consideration elsewhere, and all authors have approved this submission. The authors declare no competing interests.

Use of AI-assisted tools. Claude Code (Anthropic) and ChatGPT (OpenAI) assisted with code development and refactoring, proposals for control analyses, language editing and revision of author-written text, and the code that draws the figures; no figure or result was generated by an AI model directly. The same statement, with the tools cited, closes the manuscript.

Thank you for considering our manuscript.

Yours sincerely

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on behalf of the co-authors Minho Bae, Jinho Park, Hayoung Lee, Hyeokjin Lim and Seonghee Lee

Manuscript Information

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